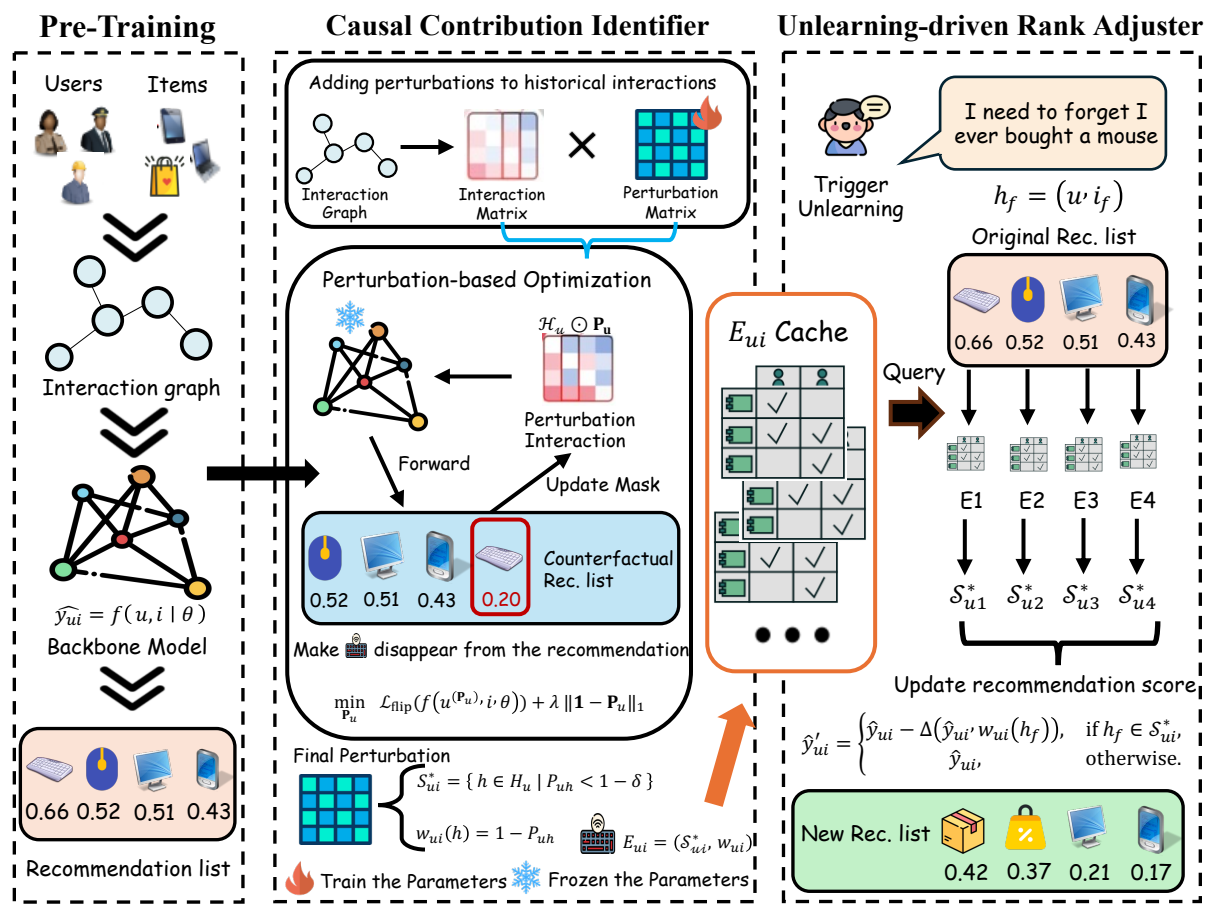

Pre-Training
Causal Contribution Identifier
Unlearning-driven Rank Adjuster
Users
Items
Interaction graph
$\widehat{y_{ui}} = f(u, i \mid \theta)$
Backbone Model
0.66 0.52 0.51 0.43
Recommendation list
Adding perturbations to historical interactions
Interaction Graph
Interaction Matrix
Perturbation Matrix
Perturbation-based Optimization
$\mathcal{H}_u \odot \mathbf{P_u}$
Perturbation Interaction
Forward
Update Mask
0.52 0.51 0.43 0.20
Counterfactual Rec. list
Make disappear from the recommendation
$\min_{\mathbf{P_u}} \mathcal{L}_{\text{flip}}(f(u^{(\mathbf{P_u})}, i, \theta)) + \lambda \|\mathbf{1} - \mathbf{P_u}\|_1$
Final Perturbation
$S^*_{ui} = \{ h \in H_u \mid P_{uh} < 1 - \delta \}$
$w_{ui}(h) = 1 - P_{uh}$
$E_{ui} = (S^*_{ui}, w_{ui})$
Train the Parameters
Frozen the Parameters
$E_{ui}$ Cache
Trigger Unlearning
I need to forget I ever bought a mouse
$h_f = (u, i_f)$
Original Rec. list
0.66 0.52 0.51 0.43
Query
E1 E2 E3 E4
$S^*_{u1}$ $S^*_{u2}$ $S^*_{u3}$ $S^*_{u4}$
Update recommendation score
$\hat{y}'_{ui} = \begin{cases} \hat{y}_{ui} - \Delta(\hat{y}_{ui}, w_{ui}(h_f)), & \text{if } h_f \in S^*_{ui}, \\ \hat{y}_{ui}, & \text{otherwise.} \end{cases}$
New Rec. list
0.42 0.37 0.21 0.17